HALEY PATEL

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EDUCATION

New Jersey Institute of Technology, *Albert Dorman Honor's College, Ying Wu College of Computing* *September 2021- May 2025*

Major*: Computer Science* **Minor***: Applied Mathematics* **Cumulative GPA: 3.916/4.00**

- Received full honors merit scholarship for all years of attendance. Committed 30+ external volunteer hours/semester.
- Relevant coursework: Java, C/C++, Python, Data Structures and Algorithms, Database Systems, Quantitative Analysis with Business Problems, Statistics, Multivariable Calculus, Actuarial Mathematics, Finance

PROJECTS

Honor's Summer Research Institute: Machine Learning for Advanced Manufacturing Process Optimization *Current*

- Applying math knowledge to develop an iterative machine learning algorithm, using RNN and LTSM deep learning models, to simulate the heat profiles of particles in advanced manufacturing processes and optimize heat manufacturing.
- Utilized the Anaconda environment for the use of Python Numpy, Pandas, Scikit-Learn, and PyTorch libraries to process 9 million datapoints for the heat and spaciotemporal profiles of particles manufactured in a laser bed.

To-do List Application built with RESTful API (PostgreSQL, Node.js , Express, React)

- Developed full-stack, user-friendly, to-do list application by developing a RESTful API using Node.js and Express to enable seamless communication between the front-end and the server. Implemented endpoints for CRUD operations on tasks.
- Utilized PostgreSQL to store and manage task data and built the front end with React, Bootstrap, and HTML.

Bank Database and Website Project (SQL, PHP, MySQL, HTML, CSS) *February – April 2023*

- Conceptualized and built a database on Oracle SQL Server to replicate the requirements for a fictional banking institution.
- Utilized PHP and MySQL to establish the backend infrastructure for the operational banking website. Designed with HTML and CSS, the website includes login pages, customer transaction features, and an employee management dashboard.

RSA Encryption Algorithm Implementation (Python) *May 2023*

- Programmed a discrete mathematics based implementation of the RSA encryption algorithm.
- Improved modular arithmetic functions with bitwise shifting to decrease the encryption and decryption time and computational cost of exponentiating extremely large numbers with each other significantly.

PROFESSIONAL EXPERIENCES

Software Development Intern, UPS (United Parcel Service) *(Current)* ***June 2023 – August 2023***

- Assigned to develop and deploy a full-stack employee email server management tool for internal use to increase the efficiency of migrating employee email servers across regions. Created a Jira and Confluence board to organize objectives, and track development progress. Languages: C#, C++, HTML, CSS, React, Node.js, Kubernetes
- Spearheading an employee user experience enhancement project by gathering and analyzing usage data to pinpoint inefficiencies or confusions in employee workflows to improve overall employee user experience with enterprise tools.

Data and Claims Intern, Plymouth Rock Assurance **June 2022 – August 2022**

- Utilized the insurance department's data management systems to support data entry and analysis tasks, increasing personal efficiency in resolving claims by 40% by the end of the internship.

EXTRACURRICULAR EXPERIENCES

***Vice President*, Business Masterminds NJIT** *January 2022 - Present*

- Hosted Project Management Workshops to introduce students to SCRUM/Agile methodologies. Maintained connections with 10+ industry professionals and school alumni to serve as guest speakers in workshops and networking events.
- Initiated a collaboration with campus technology organizations to host an Artificial Intelligence in Business Workshop of 50+ attendees to teach how to leverage artificial intelligence tools to solve common business problems.

***Financial Analyst* – Investment Fund NJIT** *September 2021 - Present*

- Analyzed various publicly traded companies found through conducting industry/market research to find viable long term growth investment opportunities to increase the value of the club's stock portfolio.

Other Experiences: Women in Computing Society member, Spanish Honor Society

SKILLS

Technology: Java, C++, C, Python, HTML, CSS, React, JavaScript, Node.js, SQL, MySQL, PostgreSQL, Git, PHP, RESTful APIs, Express

Extra Skills: Creative arts, 3D modeling, Gujurati Language

Certifications: New Jersey State Seal of Biliteracy in Spanish